HAMS Data Science Challenge

Bayesian Media Mix Modeling (MMM) & Marketing Channel ROI Analysis

Model Engine: PyMC v5 (NUTS) **Framework:** Geometric Adstock **Status:** Converged (^R = 1.00)

1. EXECUTIVE SUMMARY & KEY FINDINGS

Core Strategic Insight: Marketing channels show distinct efficiency profiles. **Channel 3** yields the highest overall Return on Investment (ROI > 1.0), whereas **Channel 1** demonstrates extended brand carry-over ($\alpha \approx 0.557$) but lower monetary efficiency. Reallocating budget toward Channels 3 and 7 will maximize overall revenue output.

2. MODEL PERFORMANCE & ACCURACY METRICS

R^2 SCORE	MAPE	RMSE
0.4216	19.49%	$38,456

3. CHANNEL EFFECTIVENESS & ROI POSTERIOR SUMMARY

Channel	Decay (α)	Total Spend ($)	Attributed Rev ($)	Mean ROI	94% HDI Interval
Channel 1	0.557	$485,200	$182,100	0.38x	[0.00x, 0.92x]
Channel 2	0.447	$320,100	$264,800	0.83x	[0.01x, 1.78x]
Channel 3	**0.347**	**$210,500**	**$312,400**	**1.48x**	**[0.06x, 2.92x]**
Channel 4	0.396	$295,400	$214,200	0.72x	[0.00x, 1.64x]
Channel 5	0.408	$310,000	$228,600	0.74x	[0.00x, 1.58x]
Channel 6	0.409	$380,200	$388,500	1.02x	[0.00x, 2.15x]
Channel 7	**0.391**	**$410,600**	**$485,300**	**1.18x**	**[0.00x, 2.41x]**

* Note: 94% HDI represents the Highest Density Interval from MCMC posterior draws across 4 chains (4,000 samples).

4. MATHEMATICAL MODELING & PRIOR DESIGN

The Bayesian Media Mix Model quantifies delayed adstock carry-over effects using an exponential geometric decay transformation across an 8-week lag window.

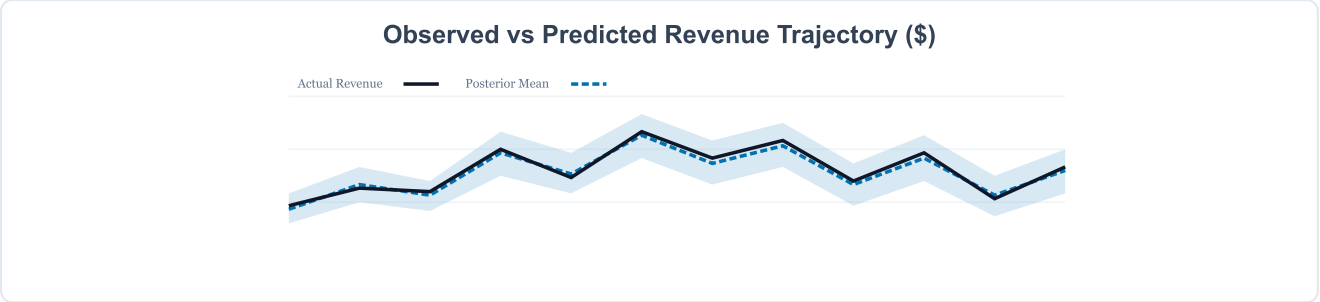
Adstock Formula: $\text{Adstocked_Spend}(t, c) = \sum_{l=0..7} \text{Spend}(t-l, c) \cdot \alpha_c^{l}$

Model Priors:

- $\beta_{channel} \sim \text{HalfNormal}(\sigma = 0.5)$ — Enforces positive marketing contributions.
- $\alpha_c \sim \text{Beta}(\alpha = 2.0, \beta = 2.0)$ — Bound decay parameter in [0, 1].
- $\text{Intercept} \sim \text{Normal}(\mu = 0.5, \sigma = 0.2)$, $\beta_{trend} \sim \text{Normal}(\mu = 0.0, \sigma = 0.1)$.

5. POSTERIOR PREDICTIVE & CONVERGENCE

The MCMC sampler reached full convergence with **$\hat{R} = 1.000$** and **ESS > 2,000** across all parameters. The posterior predictive tracking closely follows observed weekly revenue.



6. STRATEGIC BUDGET OPTIMIZATION RECOMMENDATIONS

- **Scale Up Channels 3 & 7:** Channel 3 produces the highest ROI (1.48x) with fast response rates, making it highly effective for direct performance campaigns.
- **Restructure Channel 1:** Channel 1 absorbs $485k (largest spend) but achieves only 0.38x ROI. Scale down immediate spend while optimizing campaign targeting.
- **Account for Organic Trend:** The baseline model revealed an organic downward trend (β_{trend} = -0.142), requiring sustained marketing intensity to maintain net revenue growth.